COURSE: CS 351 — Machine Learning Foundations

Credits: 3 **|** Department: Computer Science

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PREREQUISITES: CS 301 (minimum grade C) AND MATH 241 AND MATH 251

CO-REQUISITES: None

OFFERED: Fall, Spring

DESCRIPTION:

Mathematical foundations of machine learning. Topics include supervised learning (linear/logistic regression, SVMs, decision trees, ensemble methods), unsupervised learning (k-means, PCA, autoencoders), bias-variance tradeoff, regularization (L1/L2), cross-validation, evaluation metrics, and an introduction to neural networks. Heavy use of Python (NumPy, scikit-learn, PyTorch intro).

MINIMUM GRADE TO SATISFY PREREQUISITE FOR CS 352, CS 353, CS 354, CS 355, CS 356: No minimum specified beyond completion (C or above implied by standard policy)

GRADING: Problem sets 30%, Programming assignments 40%, Midterm 15%, Final 15%.

NOTE: Students must be comfortable with linear algebra (matrix operations, eigenvalues) and probability (Bayes theorem, expectation) before enrolling. Self-assessment quiz available on course page.